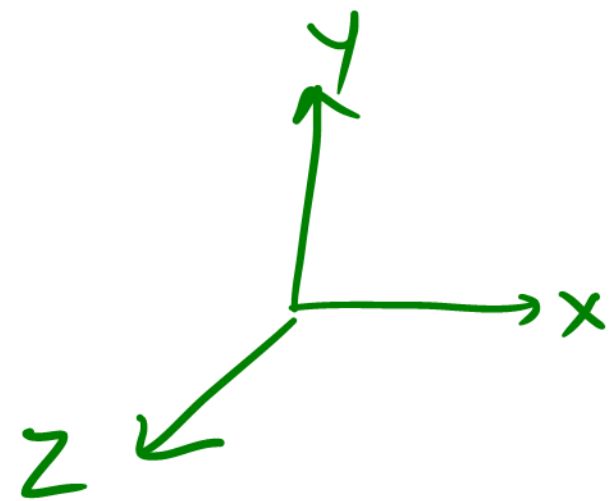
Principles of Statics

$\sum \bar{F} = 0$ then the particle has zero linear acceleration

$\sum \bar{M} = 0$ then the body has zero angular acceleration

In terms of components

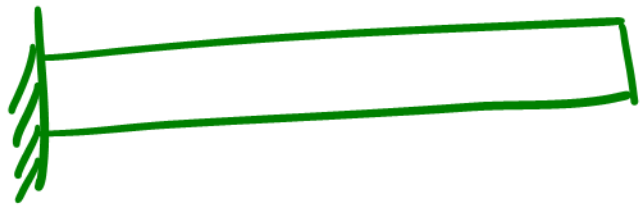
$$\sum F_x = 0 \qquad \sum M_x = 0$$
$$\sum F_y = 0 \qquad \sum M_y = 0$$
$$\sum F_z = 0 \qquad \sum M_z = 0$$

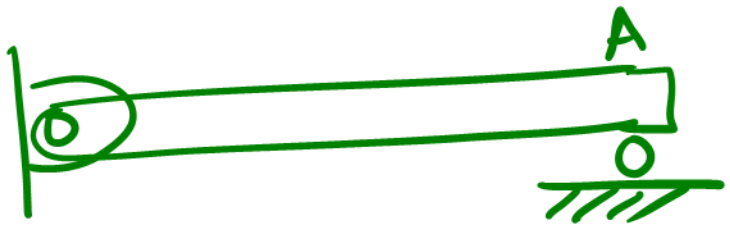


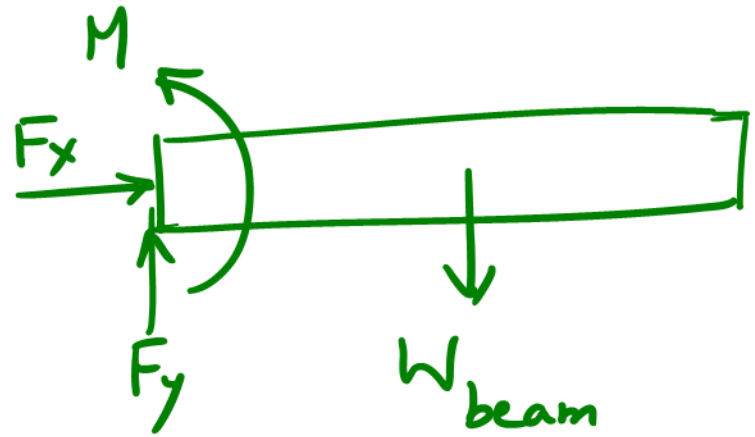
Free body diagram

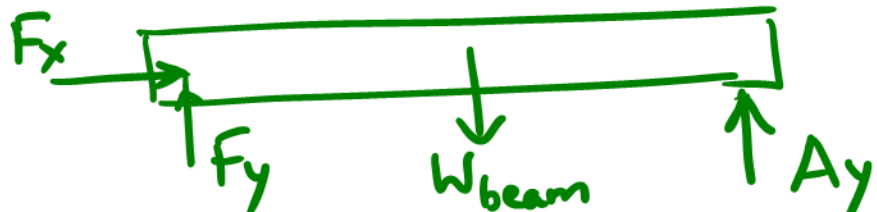
Isolate a body

Represent the effects of its surroundings by means of forces & moments









Forces involved in the human body

INTERNAL

Muscles, ligaments, joint forces

unknowns

EXTERNAL

gravity, manual or mechanical forces applied during different activities, exercise, therapy, forces applied by assistive devices

To do a mechanical analysis, we need to know

- Proper locations of muscle attachments
- Line of action of muscle forces
- masses of the body segments and locations of the centres of gravity
- type of joint
- anatomical axes of rotation of the joint

BSP - body segment parameters

Assumptions for analysis

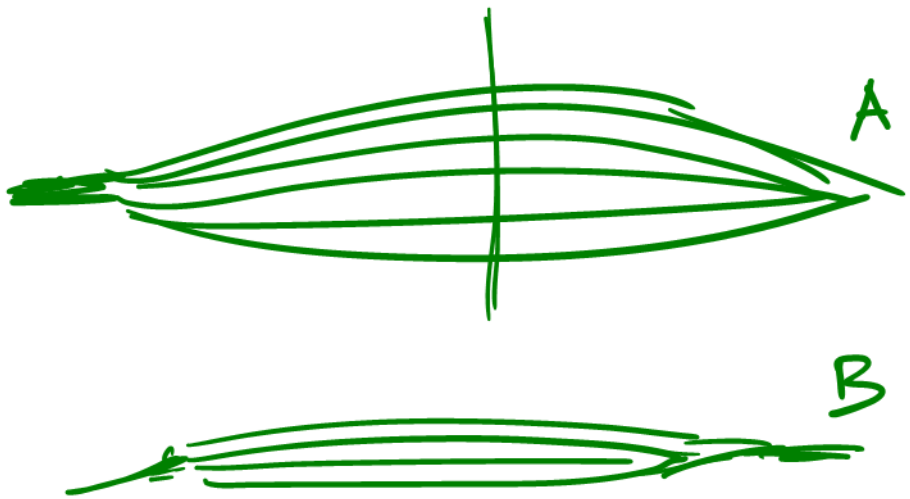
Statics problems

- Planar 2D analysis
- Inertial effects ignored
- Frictional forces @ the joints are negligible
- Segmental masses & CGs are known
- Effect of ligaments, tendons etc neglected
- Locations of muscle attachments are known
- Lines of action of muscle tension
- Anatomical axes of joints are known

2D analysis

$$\left.\begin{aligned} \sum F_x &= 0 \\ \sum F_y &= 0 \\ \sum M_z &= 0 \end{aligned}\right\}$$ 3 scalar equations

3 unknowns can be solved for



Muscle force F_m

Joint reactions J_x, J_y

PCSA

We assume anthropometric data about the segment to be analyzed is available

Muscles → cables

Bones → beams / rigid bodies

Human joints → mechanical joints

FBD

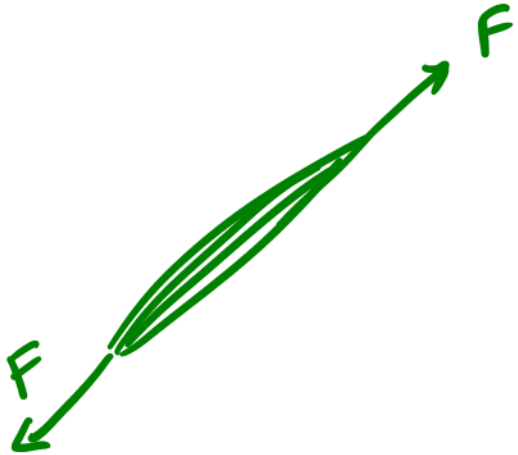
Unknown

Muscle
(Flexible member like cable)



Mag. of the tension





Mag. of the tension

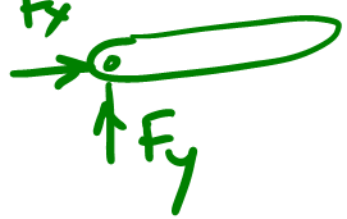
Bone on bone contact



Force along the common normal of the surfaces in contact

Mag. of force

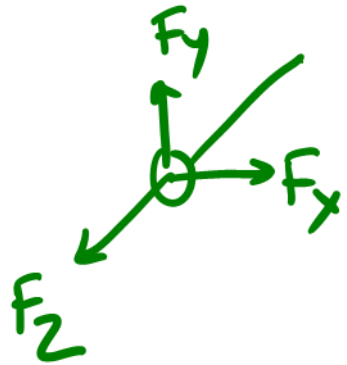
Hinge or pin connections



Elbow
Knee

2 unknowns
F_x, F_y
or mag. & dirn. of force

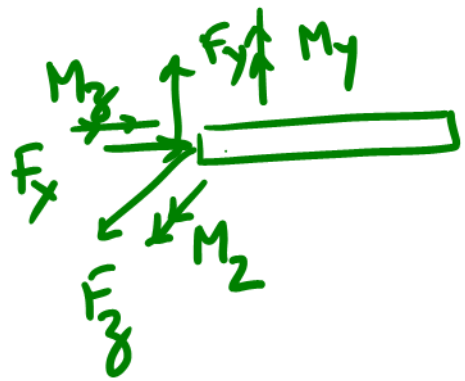
Ball & socket joint



3 unknowns

F_x, F_y, F_z

Fixed, welded or built in



6 unknowns

3 forces, 3 moments